

UNIVERSIDAD DE O'HIGGINS

Institute of Engineering Sciences
Mathematical Methods in Bioengineering

MOCK FINAL EXAMINATION

Format: Comprehensive 4-Hour Exam — **Professor:** Duvan Henao — **Date:** July 2026

General Instructions:

- **Duration:** 4 Hours (240 minutes).
- **Structure:** This exam consists of 3 comprehensive, multi-part problems covering Thermal PDEs, Lagrange Multipliers, and Chi-Squared Theory/Tests.
- **Rigor Requirement:** Show all steps clearly. For variable substitutions in integrals, explicitly state transformations and limits. Hybrid or incomplete variable changes will incur penalties.
- **Interpretation:** Complete physical/biological interpretations where requested.

Problem 1: Thermal Conduction in a Microfluidic Scaffold (Thermal Equation)

In tissue engineering, photothermal stimulation is used to induce localized cellular responses inside polymeric scaffolds. Consider a thin, uniform cylindrical hydrogel scaffold of length $L = \pi$ cm aligned along the x -axis from $x = 0$ to $x = \pi$.

The spatial-temporal temperature distribution $u(x, t)$ (in °C above ambient temperature) is modeled by the 1D heat equation:

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}, \quad 0 < x < \pi, \quad t > 0$$

where $k = 1 \text{ cm}^2/\text{s}$ is the thermal diffusivity of the hydrogel substrate. Both ends of the scaffold are maintained in temperature-controlled microfluidic channels kept at ambient temperature:

$$u(0, t) = 0 \quad \text{and} \quad u(\pi, t) = 0 \quad \text{for all } t > 0.$$

At time $t = 0$, a laser pulse creates an initial non-uniform heat distribution given by:

$$u(x, 0) = 50 \sin(x) - 20 \sin(3x).$$

- State the separation of variables ansatz $u(x, t) = X(x)T(t)$. Derive the associated spatial boundary value problem for $X(x)$ and find all valid eigenvalues λ_n and eigenfunctions $X_n(x)$ satisfying $X(0) = X(\pi) = 0$.
- Derive the general solution $u(x, t)$ using Fourier series principles and apply the initial condition $u(x, 0)$ to calculate the exact analytic expression for $u(x, t)$.
- Determine the temperature at the midpoint of the scaffold ($x = \pi/2$) as a function of time t . At what exact time t^* will the midpoint temperature decay to 10°C ? (*Leave your answer in terms of natural logarithms*).

- (d) Suppose the experiment is modified such that the boundary at $x = \pi$ is fully insulated ($\frac{\partial u}{\partial x}(\pi, t) = 0$) while $x = 0$ remains at $u(0, t) = 0$. Explain how this boundary change alters the steady-state solution $u_s(x) = \lim_{t \rightarrow \infty} u(x, t)$ and give the physical reason for this behavior.

Problem 2: Biomechanical Energy Optimization (Lagrange Multipliers)

A bioengineer is designing an extracellular matrix (ECM) scaffold construct. The total strain energy density $E(x, y, z)$ of the matrix under dynamic mechanical loading depends on three structural fiber orientation parameters $x, y, z > 0$ according to the function:

$$E(x, y, z) = x^2 + 2y^2 + 3z^2.$$

To ensure structural integrity and nutrient porosity under mechanical compression, the parameters must satisfy a linear structural constraint:

$$g(x, y, z) = x + 2y + z = 18.$$

- (a) Define the Lagrangian function $\mathcal{L}(x, y, z, \lambda)$ for this constrained optimization problem. Write down the complete vector system of equations given by $\nabla \mathcal{L} = \mathbf{0}$.
- (b) Solve the system of equations derived in part (a) to find the unique critical point (x^*, y^*, z^*) and the corresponding Lagrange multiplier λ^* .
- (c) Evaluate the minimum strain energy density $E^* = E(x^*, y^*, z^*)$. Show that any other point satisfying the constraint yields a strictly greater energy density.
- (d) Suppose the structural constraint is relaxed from $g(x, y, z) = 18$ to $g(x, y, z) = 18 + \Delta c$, defining the optimal minimum energy as a function $E^*(c)$ where $c = 18$. Prove mathematically that:

$$\frac{dE^*}{dc} = \lambda^*.$$

Provide a biological and physical interpretation of what λ^* represents to a biomechanical engineer.

Problem 3: Cell Adhesion Phenotyping (Chi-Squared Tests & Theory)

To test the biocompatibility of a surface micro-patterning technique, tissue engineers seeded mesenchymal stem cells onto two types of hydrogels: **Control Hydrogels** (standard surface) and **Functionalized Hydrogels** (patterned with cell-adhesive peptides).

After 24 hours, the cells were evaluated for successful cell spreading and focal adhesion formation. The observed cell counts are summarized in the following 2×2 contingency table:

Surface Type	Attached & Spread (O_1)	Unattached / Round (O_2)	Total
Control Hydrogel	60	40	100
Functionalized Hydrogel	80	20	100
Total	140	60	200

- (a) Let $Z_1, Z_2, \dots, Z_k$ be independent standard normal random variables ($Z_i \sim N(0, 1)$). By definition, the variable $Q = \sum_{i=1}^k Z_i^2$ follows a Chi-squared distribution with k degrees of freedom ($\chi^2(k)$). Using the properties of expected values for normal distributions, mathematically prove that:

$$\mathbb{E}[Q] = k.$$

- (b) Formulate the null hypothesis H_0 and alternative hypothesis H_1 regarding the independence of cell adhesion and hydrogel functionalization. Calculate the table of expected cell frequencies E_{ij} under H_0 .
- (c) Calculate the Chi-squared test statistic:

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}}.$$

State the number of degrees of freedom $df = (r-1)(c-1)$ and explain why $df = 1$ in this context.

- (d) Using a significance level of $\alpha = 0.05$ and the critical value $\chi_{\text{crit}}^2 = 3.841$ for $df = 1$, state your statistical decision. What is the practical bioengineering conclusion regarding the surface functionalization?

COMPLETE SOLUTIONS & MARKING SCHEME

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Solution to Problem 1: Thermal Conduction in Scaffold

Part (a): Separation of Variables & Sturm-Liouville Problem

- Let $u(x, t) = X(x)T(t)$. Substituting into $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ ($k = 1$):

$$X(x)T'(t) = X''(x)T(t) \implies \frac{T'(t)}{T(t)} = \frac{X''(x)}{X(x)} = -\lambda \quad (\text{constant}).$$

- Spatial ODE: $X''(x) + \lambda X(x) = 0$ with $X(0) = 0$ and $X(\pi) = 0$.
- Case $\lambda \leq 0$ yields only trivial solutions $X(x) = 0$.
- For $\lambda > 0$, set $\lambda = \beta^2$. General solution: $X(x) = A \cos(\beta x) + B \sin(\beta x)$.
- Boundary conditions:

$$X(0) = A = 0, \quad X(\pi) = B \sin(\beta\pi) = 0 \implies \beta_n = n \quad (n = 1, 2, 3, \dots).$$

- Eigenvalues: $\lambda_n = n^2$, Eigenfunctions: $X_n(x) = \sin(nx)$.

Part (b): General Solution & Fourier Coefficients

- Temporal ODE: $T'_n(t) + n^2 T_n(t) = 0 \implies T_n(t) = C_n e^{-n^2 t}$.
- Superposition yields:

$$u(x, t) = \sum_{n=1}^{\infty} C_n e^{-n^2 t} \sin(nx).$$

- Applying $u(x, 0) = 50 \sin(x) - 20 \sin(3x)$:

$$\sum_{n=1}^{\infty} C_n \sin(nx) = 50 \sin(x) - 20 \sin(3x).$$

- Equating coefficients: $C_1 = 50$, $C_3 = -20$, and $C_n = 0$ for $n \neq 1, 3$.
- Exact Analytic Solution:

$$u(x, t) = 50e^{-t} \sin(x) - 20e^{-9t} \sin(3x).$$

Part (c): Midpoint Temperature & Decay Time

- Evaluate at $x = \pi/2$: $\sin(\pi/2) = 1$, $\sin(3\pi/2) = -1$.

$$u\left(\frac{\pi}{2}, t\right) = 50e^{-t}(1) - 20e^{-9t}(-1) = 50e^{-t} + 20e^{-9t}.$$

- Set $u(\pi/2, t^*) = 10 \implies 50e^{-t^*} + 20e^{-9t^*} = 10$.
- Since $e^{-9t^*} \ll e^{-t^*}$ as t grows, $50e^{-t^*} \approx 10 \implies e^{-t^*} = \frac{1}{5}$.
- Taking natural logarithm: $t^* = \ln(5)$ seconds ≈ 1.6094 s.

Part (d): Insulated Boundary Physical Interpretation

- If $\frac{\partial u}{\partial x}(\pi, t) = 0$, heat flux across $x = \pi$ is zero. However, $x = 0$ remains fixed at ambient temperature $u(0, t) = 0$.
- Heat drains continuously through $x = 0$. Without continuous internal volumetric heat sources ($f(x) = 0$), all thermal energy inside the scaffold eventually escapes through $x = 0$.
- Steady-state solution remains $\mathbf{u}_s(\mathbf{x}) = \lim_{t \rightarrow \infty} \mathbf{u}(\mathbf{x}, t) = \mathbf{0}$.

Solution to Problem 2: Biomechanical Energy Optimization

Part (a): Lagrangian Formulation

- Objective: $E(x, y, z) = x^2 + 2y^2 + 3z^2$, Constraint: $g(x, y, z) = x + 2y + z = 18$.

- Lagrangian:

$$\mathcal{L}(x, y, z, \lambda) = x^2 + 2y^2 + 3z^2 - \lambda(x + 2y + z - 18).$$

- Stationarity conditions $\nabla \mathcal{L} = \mathbf{0}$:

$$\frac{\partial \mathcal{L}}{\partial x} = 2x - \lambda = 0 \implies x = \frac{\lambda}{2}$$

$$\frac{\partial \mathcal{L}}{\partial y} = 4y - 2\lambda = 0 \implies y = \frac{\lambda}{2}$$

$$\frac{\partial \mathcal{L}}{\partial z} = 6z - \lambda = 0 \implies z = \frac{\lambda}{6}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = -(x + 2y + z - 18) = 0.$$

Part (b): Solving the System

- Substitute expressions for x, y, z into the constraint:

$$\left(\frac{\lambda}{2}\right) + 2\left(\frac{\lambda}{2}\right) + \left(\frac{\lambda}{6}\right) = 18 \implies \frac{3\lambda + 6\lambda + \lambda}{6} = 18 \implies \frac{10\lambda}{6} = 18.$$

- Solving for λ^* : $\lambda^* = \frac{108}{10} = 10.8$.
- Critical point coordinates:

$$\mathbf{x}^* = 5.4, \quad \mathbf{y}^* = 5.4, \quad \mathbf{z}^* = 1.8.$$

Part (c): Minimal Energy Calculation

- Minimum strain energy density:

$$E^* = (5.4)^2 + 2(5.4)^2 + 3(1.8)^2 = 29.16 + 2(29.16) + 3(3.24) = 29.16 + 58.32 + 9.72 = \mathbf{97.2}.$$

- Since $E(x, y, z)$ is strictly convex, (x^*, y^*, z^*) is the global minimum on the constraint plane.

Part (d): Mathematical Proof & Shadow Price Interpretation

- For general c , $x(c) = \frac{3c}{10}, y(c) = \frac{3c}{10}, z(c) = \frac{c}{10}$.
- Substituting into $E^*(c)$:

$$E^*(c) = \left(\frac{3c}{10}\right)^2 + 2\left(\frac{3c}{10}\right)^2 + 3\left(\frac{c}{10}\right)^2 = \frac{9c^2 + 18c^2 + 3c^2}{100} = 0.3c^2.$$

- Differentiating with respect to c :

$$\frac{dE^*}{dc} = 0.6c \implies \left. \frac{dE^*}{dc} \right|_{c=18} = 0.6(18) = \mathbf{10.8} = \lambda^*. \quad \square$$

- Interpretation:** λ^* is the *sensitivity* or *shadow price* of the energy density relative to the structural constraint capacity c . Increasing c by 1 unit increases minimal strain energy by 10.8 units.

Solution to Problem 3: Cell Adhesion Phenotyping

Part (a): Theoretical Expectation Proof

- For $Z_i \sim N(0, 1)$, mean $\mathbb{E}[Z_i] = 0$ and variance $\text{Var}(Z_i) = 1$.
- By definition of variance: $\text{Var}(Z_i) = \mathbb{E}[Z_i^2] - (\mathbb{E}[Z_i])^2 \implies 1 = \mathbb{E}[Z_i^2] - 0 \implies \mathbb{E}[Z_i^2] = 1$.
- By linearity of expectation:

$$\mathbb{E}[Q] = \mathbb{E}\left[\sum_{i=1}^k Z_i^2\right] = \sum_{i=1}^k \mathbb{E}[Z_i^2] = \sum_{i=1}^k 1 = \mathbf{k}. \quad \square$$

Part (b): Hypotheses & Expected Frequency Table

- H_0 : Cell adhesion outcome is independent of hydrogel surface functionalization.
- H_1 : Cell adhesion outcome is dependent on hydrogel surface functionalization.
- Formula: $E_{ij} = \frac{(\text{Row } i \text{ Total}) \times (\text{Column } j \text{ Total})}{\text{Grand Total}}$.

Expected Frequencies (E_{ij})	Attached & Spread	Unattached / Round	Total
Control Hydrogel	$E_{11} = \frac{100 \times 140}{200} = \mathbf{70}$	$E_{12} = \frac{100 \times 60}{200} = \mathbf{30}$	100
Functionalized Hydrogel	$E_{21} = \frac{100 \times 140}{200} = \mathbf{70}$	$E_{22} = \frac{100 \times 60}{200} = \mathbf{30}$	100
Total	140	60	200

Part (c): Chi-Squared Calculation & Degrees of Freedom

- Calculation:

$$\chi^2 = \frac{(60 - 70)^2}{70} + \frac{(40 - 30)^2}{30} + \frac{(80 - 70)^2}{70} + \frac{(20 - 30)^2}{30}$$

$$\chi^2 = \frac{100}{70} + \frac{100}{30} + \frac{100}{70} + \frac{100}{30} = 2 \cdot \left(\frac{10}{7} + \frac{10}{3}\right) = 2 \cdot \left(\frac{100}{21}\right) \approx \mathbf{9.5238}.$$

- Degrees of freedom: $df = (2 - 1)(2 - 1) = \mathbf{1}$. Given fixed marginal totals, choosing 1 cell uniquely fixes all 3 remaining entries.

Part (d): Statistical Decision & Bioengineering Conclusion

- Test statistic $\chi_{\text{calc}}^2 = 9.524 > \chi_{\text{crit}}^2 = 3.841$ at $\alpha = 0.05$.
- **Decision: Reject H_0 .**
- **Bioengineering Conclusion:** There is statistically significant evidence that peptide surface functionalization enhances mesenchymal stem cell adhesion and focal spreading on hydrogel scaffolds.